

DEREXI: POLICY PILOT

Public Key Cryptography

RSA key pairs • OAEP encryption • X.509 PKI • Hybrid key exchange

OBJECTIVE

This lab satisfies the public-key cryptography rubric by generating and managing a 2048-bit RSA key pair, performing RSA-OAEP encryption and decryption, creating a self-signed X.509 certificate as a simple PKI example, and protecting a random AES-256 session key with RSA. All data and keys are throwaway classroom values; production DeRexi code remains untouched.

RSA: GENERATION + ASYMMETRIC ENCRYPTION

RSA + OAEP IMPLEMENTATION

```
private_key = rsa.generate_private_key(
    public_exponent=65537, key_size=2048
)
public_key = private_key.public_key()

ciphertext = public_key.encrypt(
    plaintext,
    padding.OAEP(
        mgf=padding.MGF1(hashes.SHA256()),
        algorithm=hashes.SHA256(), label=None
    )
)
recovered = private_key.decrypt(ciphertext, oaep_padding())
```

01 Key pair

2048-bit RSA uses public exponent 65537. Private material stays secret; only metadata and a public-key fingerprint are shown.

02 OAEP

RSA encryption uses OAEP with SHA-256. OAEP adds randomized padding and avoids insecure textbook RSA behavior.

03 Verification

The recovered plaintext is compared with the original message, confirming the public-encrypt/private-decrypt round trip.

SECURITY INTERPRETATION

RSA provides confidentiality only when the private key remains secret and encryption uses a secure padding scheme. The lab deliberately avoids printing or permanently storing the private key. RSA is also treated as a small-payload primitive: it protects keys or short values rather than bulk data.

SIMPLE PKI ENVIRONMENT

The lab creates a self-signed X.509 certificate for derexi-demo.local / Aurum Capital Bank, including subject, issuer, serial number, validity period, and the RSA public key. It verifies that the certificate carries the same public key generated earlier and validates the self-signature. Because the certificate is self-signed, it is suitable for demonstration but is not automatically trusted by browsers or external systems.

PAGE 1 RESULT

RSA KEY PAIR: PASS • RSA-OAEP: PASS • X.509 PKI: PASS

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Hybrid Key Exchange & Key Management

RSA-wrapped AES-256 • AES-GCM proof • temporary-key hygiene

SECURE KEY EXCHANGE

A fresh 256-bit AES session key is generated with the operating system CSPRNG, wrapped with the RSA public key, and recovered with the matching private key. The recovered key is then used with AES-GCM to encrypt and decrypt a small DeRexi payload, demonstrating a complete hybrid-cryptography flow.

HYBRID KEY-EXCHANGE EVIDENCE

```
session_key = os.urandom(32)
wrapped_key = public_key.encrypt(
    session_key, oaep_padding()
)
recovered_key = private_key.decrypt(
    wrapped_key, oaep_padding()
)

aesgcm = AESGCM(recovered_key)
sealed = aesgcm.encrypt nonce, message, None
opened = aesgcm.decrypt nonce, sealed, None
```

RSA KEY-MANAGEMENT EVIDENCE

```
Production guidance:
- keep private keys out of Git
- store keys in KMS/HSM/secrets manager
- restrict disk permissions
- rotate and revoke when needed
- renew/revoke certificates

Demo PEM file: chmod 600
Reloaded + verified, then deleted
```

EXECUTION EVIDENCE

+ Thought: 9.1s
Lab complete, all green. Here's the summary you asked for:
1. Rubric requirements - ALL PASSED

#	Requirement	Result
1	RSA key pair generation - 2048-bit, exponent 65537, public key derived, all four labels printed, private material never printed (only fingerprint/metadata)	PASS
2	Basic asymmetric encryption - fictional message, OAEP + SHA-256, plaintext + hex ciphertext shown, decrypt + verify, all four explanations	PASS
3	Simple PKI environment - self-signed X.509 cert, CN derexi-demo.local, O Arrus Capital Bank, subject/issuer/serial/validity/public key displayed, cert public key verified — generated public key	PASS
4	Secure key exchange - random AES-256 session key + RSA wrap + unwrap + exact match, plus an AES-GCM round trip proving the hybrid flow, five explanations	PASS
5	Key management - all 7 bullets; temp PEM written to lab folder with chmod 600, reloaded, then deleted (0 files remain)	PASS
6	Terminal output - all 8 headings exact, 5 PASS lines exact, completion line exact, exit 0	PASS
7	Security considerations - all four groups (RSA / private keys / PKI / hybrid) present	PASS

01 RSA

Key generation and asymmetric encryption both returned PASS.

02 PKI + exchange

Self-signed certificate and RSA-wrapped AES key both returned PASS.

03 Safety

Key-management check passed with zero private-key leakage and no leftover demo PEM files.

RUBRIC COVERAGE

OVERALL RESULT

ALL PUBLIC-KEY CRYPTOGRAPHY RUBRIC REQUIREMENTS: MET / PASS